

AKSHAT GUPTA

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EDUCATION

IIT Bhilai

B.Tech. DSAI – CGPA: 7.86/10

Aug. 2023 – Present

Chhattisgarh, India

Douglas Memorial Higher Secondary School

ISC – **95.5%** | ICSE – **97%**

Apr. 2010 – Apr. 2022

West Bengal, India

COURSEWORK

- Data Analytics & Visualization
- Machine Learning
- Deep Learning
- Database Management Systems

PROJECTS

Multi-Agent Research Assistant | Python, LangGraph, LangChain, FastAPI, Next.js

- Designed a multi-agent system with an orchestrator-worker architecture on LangGraph, coordinating 4 specialized agents (Web Researcher, Document Analyst, Synthesizer) through shared graph state with conditional routing, Pydantic-structured routing decisions, and an iteration guard capping the plan-execute-evaluate loop at 3 cycles to guarantee termination.
- Built manual tool-calling loops (finer-grained control than a stock AgentExecutor) giving agents autonomous access to DuckDuckGo web search and PDF/text document analysis tools; served the backend on FastAPI with real-time agent-step streaming over Server-Sent Events (SSE) to a Next.js 16 frontend.

HireLens: Resume-to-Job Matching | Python, Sentence Transformers, PyTorch, Docker, AWS EC2

- Fine-tuned **all-MiniLM-L6-v2** using **CosineSimilarityLoss** with weak supervision on 5,692 resume-job pairs built from 3.3M LinkedIn postings and 19K resumes; improved Pearson cosine correlation by **+75%** ($0.355 \rightarrow 0.622$) over the base model. Designed a 4-component MatchScorer combining semantic embeddings with spaCy transformer-based NER over a 681-skill taxonomy.
- Achieved NDCG@10 **0.962**, Precision@1 **0.896**, MRR **0.948**, AUC-ROC **0.837** on 712 held-out pairs at 165 ms/resume GPU latency. Deployed full-stack (FastAPI + React + PostgreSQL + Redis + nginx) on AWS EC2 with MLflow experiment tracking, S3 model artifact storage, and a 4-job GitHub Actions CI/CD pipeline.

YouTube Video Chatbot | Python, Streamlit, LangChain, FAISS, OpenAI

- Built a RAG chatbot over YouTube transcripts with hybrid FAISS + BM25 retrieval using reciprocal-rank fusion, LLM query rewriting, and 3-variant multi-query expansion.
- Chunked transcripts into 1,000-char overlapping segments with timestamp metadata; applied LLMChainExtractor to top-6 retrieved candidates and linked answers to exact video timestamp citations.

OPEN SOURCE CONTRIBUTIONS

Cognee – AI Memory Platform for Agents | Python

- Designed a provenance-lineage subgraph for Cognee, a knowledge-graph memory engine, so every ingested node traces back to its source document and dataset, covered end-to-end by unit and integration tests in CI. [[PR](#)]
- Added video support to Cognee's ingestion pipeline by transcribing the audio track and timestamping each segment, so video content became searchable the same way text already was. [[PR](#)]

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript

Agentic AI & Orchestration: LangGraph, LangChain, Multi-Agent Systems, Tool-Calling, RAG, Vector Databases, MCP

Machine Learning & Data Science: PyTorch, Scikit-learn, spaCy, Sentence Transformers, Pandas, NumPy

MLOps: Docker, MLflow, GitHub Actions CI/CD, AWS EC2, AWS S3, nginx

Backend & API Integration: FastAPI, REST APIs, React, Next.js, Node.js

Databases: PostgreSQL, MySQL, Redis, FAISS, Chroma, MongoDB